

- Solve all the problems below in **2 hours and 30 minutes**.
- **Mobile phones are not allowed in the examination hall.**
- Please justify all your answers through proof or counterexample.
- Scoring: **1** → 6 points, **2** → 5 points, **3** → 6 points, **4** → 3 points.

1. Let be

$$f(x, y) = \begin{cases} \frac{2x^3+3y^3}{x^2+y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

(a) Show that the function f is continuous over $\mathbb{R}^2$. **(1.5 pts)**

The function f is clearly continuous on $\mathbb{R}^2 \setminus \{0\}$ as a rational fraction with nonzero denominator. Switching to polar coordinates $(x, y) = (r \cos \theta, r \sin \theta)$ we write:

$$|f(x, y)| = \left| \frac{2r^3 \cos^3 \theta + 3r^3 \sin^3 \theta}{r^2} \right| = r |2 \cos^3 \theta + 3 \sin^3 \theta| \leq 5r$$

which shows that f tends to 0 at 0 . It is therefore continuous at 0 and finally over $\mathbb{R}^2$.

(b) Compute partial derivatives of f over $\mathbb{R}^2 \setminus \{(0, 0)\}$, and at the point $(x, y) = (0, 0)$. **(2 pts)**

At $(x, y) \neq (0, 0)$ we calculate

$$\frac{\partial f}{\partial x}(x, y) = \frac{2x^4 + 6x^2y^2 - 6xy^3}{(x^2 + y^2)^2} \text{ and } \frac{\partial f}{\partial y}(x, y) = \frac{9x^2y^2 - 4x^3y + 3y^4}{(x^2 + y^2)^2}.$$

In $(0, 0)$, $\frac{f(x, 0) - f(0, 0)}{x - 0} = \frac{2x^3}{x^3} \rightarrow_{x \rightarrow 0} 2$ so $\frac{\partial f}{\partial x}(0, 0) = 2$, and we calculate the same as $\frac{\partial f}{\partial x}(0, 0) = 3$.

(c) Is the function f of class C^1 on $\mathbb{R}^2$? **(1 pt)**

We note that $\frac{\partial f}{\partial x}(0, y) = 0$ does not tend to $\frac{\partial f}{\partial x}(0, 0)$ when y tends to 0 . So f is not of class C^1 in $(0, 0)$.

(d) Study if f is differentiable at the point $(x, y) = (0, 0)$. **(1.5 pts)**

If f is differentiable at $(0, 0)$, then according to question 2. its Jacobian matrix is $(2, 3)$. Switching again to polar writing, we have

$$\frac{f(x, y) - 2x - 3y}{\|(x, y)\|} = -(2 \cos \theta \sin^2 \theta + 3 \sin \theta \cos^2 \theta)$$

which is clearly not zero bound in $r \rightarrow 0$ (set for example $\theta = \pi/4$). This contradiction shows that $f(x, y)$ is not differentiable at 0.

2. Let (S) be the system

$$\begin{cases} x \cos y + y \sin x + z = 0, \\ xe^y + ye^x + z^2 = 0. \end{cases}$$

(a) State the implicit function theorem. **(1 pt)**

See the course.

(b) Show that there exist ϵ , a real strictly positive, and a function $\phi :]-\epsilon, +\epsilon[\rightarrow \mathbb{R}^2 : z \rightarrow \phi(z) = (x, y)$ of class C^1 such that $\phi(0) = (0, 0)$ and $(x(z), y(z), z)$ is solution of the system (S). **(2.5 pts)**

Let us define the following function

$$\begin{aligned} f : \mathbb{R}^2 \times \mathbb{R} &\longrightarrow \mathbb{R} \\ ((x, y), z) &\longmapsto f(((x, y), z)) = (f_1(((x, y), z)), f_2(((x, y), z))) \\ &= (x \cos y + y \sin x + z, xe^y + ye^x + z^2) \end{aligned}$$

f is of class C^1 (each coordinate is $C^1(\mathbb{R}^3)$). One has $f(((0, 0), 0)) = 0$ and

$$D_1 f(((x, y), z)) = \begin{pmatrix} \frac{\partial f_1(((x, y), z))}{\partial x} & \frac{\partial f_1(((x, y), z))}{\partial y} \\ \frac{\partial f_2(((x, y), z))}{\partial x} & \frac{\partial f_2(((x, y), z))}{\partial y} \end{pmatrix} = \begin{pmatrix} \cos y + y \cos x & -x \sin y + \sin x \\ e^y + ye^x & xe^y + e^x \end{pmatrix}$$

and

$$D_1 f(((0, 0), 0)) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

so the determinant of $D_1 f(((0, 0), 0))$ is equal to 1, thus the Jacobian matrix at $((0, 0), 0)$ is invertible and by the implicit function theorem, there exist a neighborhood U of 0, of the form $U =]-\epsilon, +\epsilon[$, and a function $\phi :]-\epsilon, +\epsilon[\rightarrow \mathbb{R}^2 : z \mapsto \phi(z) = (x, y)$ of class C^1 such that $\phi(0) = (0, 0)$ and $f(\phi(z), z) = f((x(z), y(z)), z) = 0$, for all z in $]-\epsilon, +\epsilon[$. This means that $(x(z), y(z), z)$ is a solution of (S).

(c) Calculate $\phi'(0)$. **(1.50 pts)**

ϕ is of class C^1 , so $z \mapsto x(z)$, and $z \mapsto y(z)$ are of class C^1 and $\phi(0) = (0, 0) = (x(0), y(0))$. Indeed $\forall z \in]-\epsilon, +\epsilon[$, one has $f((x(z), y(z)), z) = 0$, equivalently, we have

$$\begin{aligned} x(z) \cos y(z) + y(z) \sin x(z) + z &= 0, \\ x(z)e^{y(z)} + y(z)e^{x(z)} + z^2 &= 0. \end{aligned}$$

Derivating both sides of the equations, we get

$$\begin{aligned} x'(z) \cos y(z) - x(z) \sin y(z) y'(z) + y'(z) \sin x(z) + y(z) \sin x(z) x'(z) + 1 &= 0, \\ (x'(z) + x(z) y'(z)) e^{y(z)} + (y'(z) + x'(z) y(z)) e^{x(z)} + 2z &= 0. \end{aligned}$$

Set $z = 0$, we obtain

$$\begin{aligned} x'(0) + 1 &= 0, \\ x'(0) + y'(0) &= 0. \end{aligned}$$

thus $\phi'(0) = (x'(0), y'(0)) = (-1, 1)$.

3. Let $\phi :]0, +\infty[\rightarrow \mathbb{R}$ be a function of class C^2 , and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be function defined by $f(x) = \phi(\|x\|)$, where $\|\cdot\|$, is the euclidean norm on $\mathbb{R}^n$.

- (a) Show that f is a function of class C^2 on $\mathbb{R}^n \setminus \{0\}$. Express its first and second differential in terms of ϕ derivatives. **(3 pts)**

The function $x \mapsto \|x\| = N_2(x)$ is of class C^2 over $\mathbb{R}^n - 0$ as a composition of two functions: $t \mapsto \sqrt{t}$ (of class C^2 on $\mathbb{R}_+^*$) and $x = (x_1, \dots, x_n) \mapsto \sum_{i=1}^n x_i^2$ (is a polynomial function of class C^2). Since both functions ϕ and N_2 are of class C^2 , then f is of class C^2 over $\mathbb{R}^n - 0$. We have $\frac{\partial N_2}{\partial x_i}(x) = \frac{x_i}{N_2(x)}$, $i = 1, \dots, n$, then the differential of N_2 at $x \in \mathbb{R}^n - 0$ is

$$DN_2(x) : h \mapsto \frac{1}{\|x\|} \sum_{i=1}^n \frac{\partial N_2}{\partial x_i}(x) h_i = \frac{\langle x, h \rangle}{\|x\|}.$$

Therefore, by chain rule, for all $h \in \mathbb{R}^n$,

$$Df(x)(h) = D(\phi \circ N_2)(x)(h) = \left\langle \phi'(\|x\|) \frac{x}{\|x\|}, h \right\rangle.$$

and the second differential is given by

$$\forall h, k \in \mathbb{R}^n, \quad D^2 f(x)(h, k) = D(Df)(x)(h)(k) = \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_j \partial x_i}(x) h_i k_j$$

where

$$\begin{aligned} \frac{\partial^2 f}{\partial x_j \partial x_i}(x) &= \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial}{\partial x_j} \left(\phi'(\|x\|) \frac{x_i}{\|x\|} \right) \\ &= \phi''(\|x\|) \frac{x_i x_j}{\|x\|^2} + \phi'(\|x\|) \left(\frac{\delta_{ij}}{\|x\|} - \frac{x_i x_j}{\|x\|^3} \right), \end{aligned}$$

$$\text{with } \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

- (b) Express $\Delta f = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}$ in term of derivatives of ϕ . **(1.5 pts)**

By previous equality, when $i = j$, we obtain:

$$\begin{aligned}\Delta f &= \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} = \sum_{i=1}^n \left(\phi'' \frac{x_i^2}{\|x\|^2} + \left(\frac{1}{\|x\|} - \frac{x_i^2}{\|x\|^3} \right) \phi' \right) \\ &= \phi'' + \frac{n-1}{\|x\|} \phi'.\end{aligned}$$

- (c) Find all the C^2 function on $\mathbb{R}^n \setminus \{0\}$, which satisfy $\Delta f = 0$ and depend only on $\|x\|$. **(1.5 pts)**

By (b), the Laplace equation $\Delta f = 0$ is equivalent to

$$\phi'' + \frac{n-1}{\|x\|} \phi' = 0,$$

it is a second degree homogeneous differential equation in $\mathbb{R}$. The solution is given by

$$f(x) = \phi(\|x\|) = \begin{cases} c \ln(\|x\|) + c_1 & \text{if } n = 2 \\ \frac{c_2}{(2-n)\|x\|^{n-2}} + c_3 & \text{if } n \neq 2. \end{cases}$$

4. Calculate the volume of the solid enclosed by the plane xoy , the cylinder $x^2 + y^2 = x$, and the sphere $x^2 + y^2 + z^2 = 1$.

Let us denote the domain by D and the volume of the solid by V , so

$$V = \iiint_D dx dy dz$$

the domain is enclosed by:

$$\begin{cases} x^2 + y^2 + z^2 \leq 1 \\ x^2 + y^2 \leq x \\ z \geq 0. \end{cases}$$

Using the cylinder coordinates, we get

$$\begin{cases} r^2 + z^2 \leq 1 \\ r^2 \leq r \cos \theta \\ z \geq 0. \end{cases}$$

$r \in [0, 1]$, $\theta \in [-\arccos r, +\arccos r]$ and $z \in [0, \sqrt{1-r^2}]$. **(1.5 pts)**

Therefore

$$V = \int_0^1 \int_{-\arccos r}^{+\arccos r} \int_0^{\sqrt{1-r^2}} r dz d\theta dr = 2 \int_0^1 r \sqrt{1-r^2} \arccos r dr = \frac{2}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right) \text{ (1.5 pts).}$$